Privacy and data governance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete fictional data-flow and governance record.

Keep fields, purpose, roles, linking routes, retention and unresolved risks together.

Keep one habit: Recall → W1–W3/W5 → independent subgroup case → research project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Research ethics and consent

> **My remembered idea, example and correction**

Sensitivity analysis

> **My remembered idea, example and correction**

Statistical tests and assumptions

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Data governance states purpose, necessary fields, readers, locations and retention. This exercise uses fictional metadata, not personal records. Main export has coded ID, exact birth date, classroom and rare condition. Their combination may identify someone even without a name. For the explicit toy task-score purpose, unnecessary details are removed in a proposed plan.

  Separate picture analysis has participant code and score; a restricted key maps codes to identities for authorized withdrawal handling. The shared code makes records pseudonymous and linkable if the key is available. Publishing both breaks that separation. Restricted access must actually be implemented; a diagram or metadata flag does not prove it.

  Rare combinations and small groups can remain identifying through other information. A proposed retention action is not executed deletion. Under the supplied minimum-five rule, groups below five are blocked; a count of five meets only that rule, not a privacy guarantee. Any suppression or aggregation also needs checks for indirect disclosure and permitted purpose.

  No export, identity lookup, real approval or deletion is performed.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Data governance states purpose, necessary fields, readers, locations and retention. This exercise uses fictional metadata, not personal records. Main export has coded ID, exact birth date, classroom and rare condition. Their combination may identify someone even without a name. For the explicit toy task-score purpose, unnecessary details are removed in a proposed plan.

  Separate picture analysis has participant code and score; a restricted key maps codes to identities for authorized withdrawal handling. The shared code makes records pseudonymous and linkable if the key is available. Publishing both breaks that separation. Restricted access must actually be implemented; a diagram or metadata flag does not prove it.

  Rare combinations and small groups can remain identifying through other information. A proposed retention action is not executed deletion. Under the supplied minimum-five rule, groups below five are blocked; a count of five meets only that rule, not a privacy guarantee. Any suppression or aggregation also needs checks for indirect disclosure and permitted purpose.

  No export, identity lookup, real approval or deletion is performed.

- Export has coded ID, exact birth date, classroom and rare condition. Team calls it anonymous because names are removed. Show the intermediate steps and give the conclusion with its scope.

Main export field	Reason for concern	Toy-purpose action
Coded ID	May link to other records or a key	Restrict under the declared role plan
Exact birth date	Detailed identifying combination	Remove from the task-score analysis proposal
Classroom	Narrows the possible people	Remove unnecessary detail

Main export field	Reason for concern	Toy-purpose action
Rare condition	Can identify through unusual combinations	Remove from this toy-purpose proposal
No names in export	Does not remove all linkage routes	Do not label it guaranteed anonymous
Missing readers/retention evidence	Plan alone does not prove controls	Record unresolved checks; do not claim actual protection

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Data governance states purpose, necessary fields, readers, locations and retention. This exercise uses fictional metadata, not personal records. Main export has coded ID, exact birth date, classroom and rare condition. Their combination may identify someone even without a name. For the explicit toy task-score purpose, unnecessary details are removed in a proposed plan.

  Separate picture analysis has participant code and score; a restricted key maps codes to identities for authorized withdrawal handling. The shared code makes records pseudonymous and linkable if the key is available. Publishing both breaks that separation. Restricted access must actually be implemented; a diagram or metadata flag does not prove it.

  Rare combinations and small groups can remain identifying through other information. A proposed retention action is not executed deletion. Under the supplied minimum-five rule, groups below five are blocked; a count of five meets only that rule, not a privacy guarantee. Any suppression or aggregation also needs checks for indirect disclosure and permitted purpose.

  No export, identity lookup, real approval or deletion is performed.

- A report omits this qualification: A size threshold or coding scheme alone does not guarantee privacy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Data governance states purpose, necessary fields, readers, locations and retention. This exercise uses fictional metadata, not personal records. Main export has coded ID, exact birth date, classroom and rare condition. Their combination may identify someone even without a name. For the explicit toy task-score purpose, unnecessary details are removed in a proposed plan.

  Separate picture analysis has participant code and score; a restricted key maps codes to identities for authorized withdrawal handling. The shared code makes records pseudonymous and linkable if the key is available. Publishing both breaks that separation. Restricted access must actually be implemented; a diagram or metadata flag does not prove it.

  Rare combinations and small groups can remain identifying through other information. A proposed retention action is not executed deletion. Under the supplied minimum-five rule, groups below five are blocked; a count of five meets only that rule, not a privacy guarantee. Any suppression or aggregation also needs checks for indirect disclosure and permitted purpose.

  No export, identity lookup, real approval or deletion is performed.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than 5. Decide.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled privacy and **data governance evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of privacy and **data governance**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Privacy and **data governance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Privacy and **data governance**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Aggregate contains one member in a subgroup; policy forbids subgroup reports smaller than 5. Decide.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

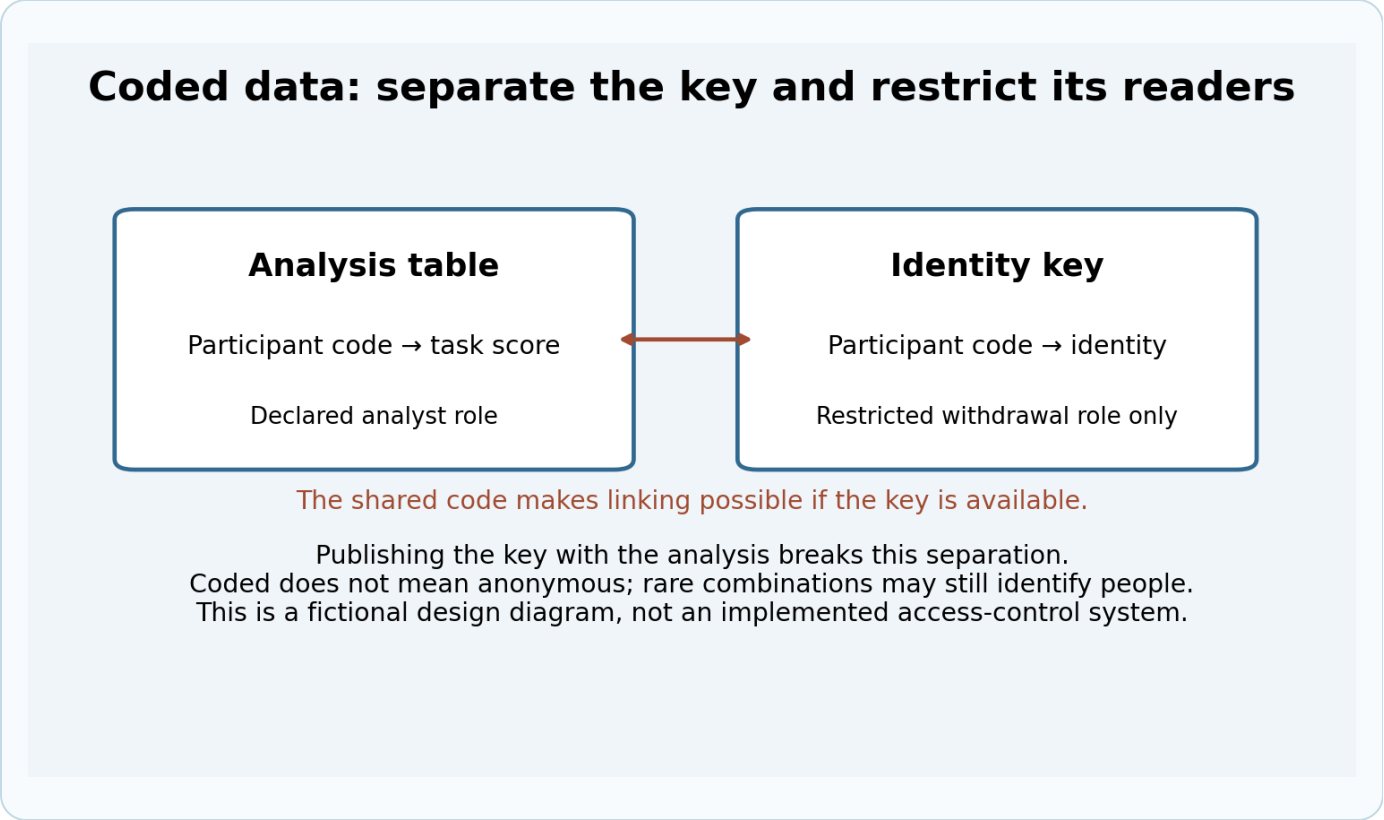
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.
- The analysis table holds codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling.
- Point to the analysis table and the restricted identity key. Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/20

Privacy and data governance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Separate tables

Keep analysis records apart from the identity key.



2. Limit access

Analysts receive scores and codes, not the name linkage.



3. Define retention

Apply the approved deletion schedule to both stores and listed copies.

CHANGE ONE THING

Publish the identity key with the analysis table.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names.
- Which labelled step matches this detail? Apply the approved deletion schedule to both stores and listed copies.
- 1. Define retention
- 2. Separate tables
- 3. Limit access
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Define retention Apply the approved deletion schedule to both stores and listed copies.

Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Changed-case question and optional hint

- Changed-case practice: Publish the identity key with the analysis table. Which conclusion is supported by the supplied facts?
- 1. The records become directly linkable to names and breach this access design.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The records become directly linkable to names and breach this access design. Separation only helps while the linkage key remains restricted.

Repair and retain the record

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

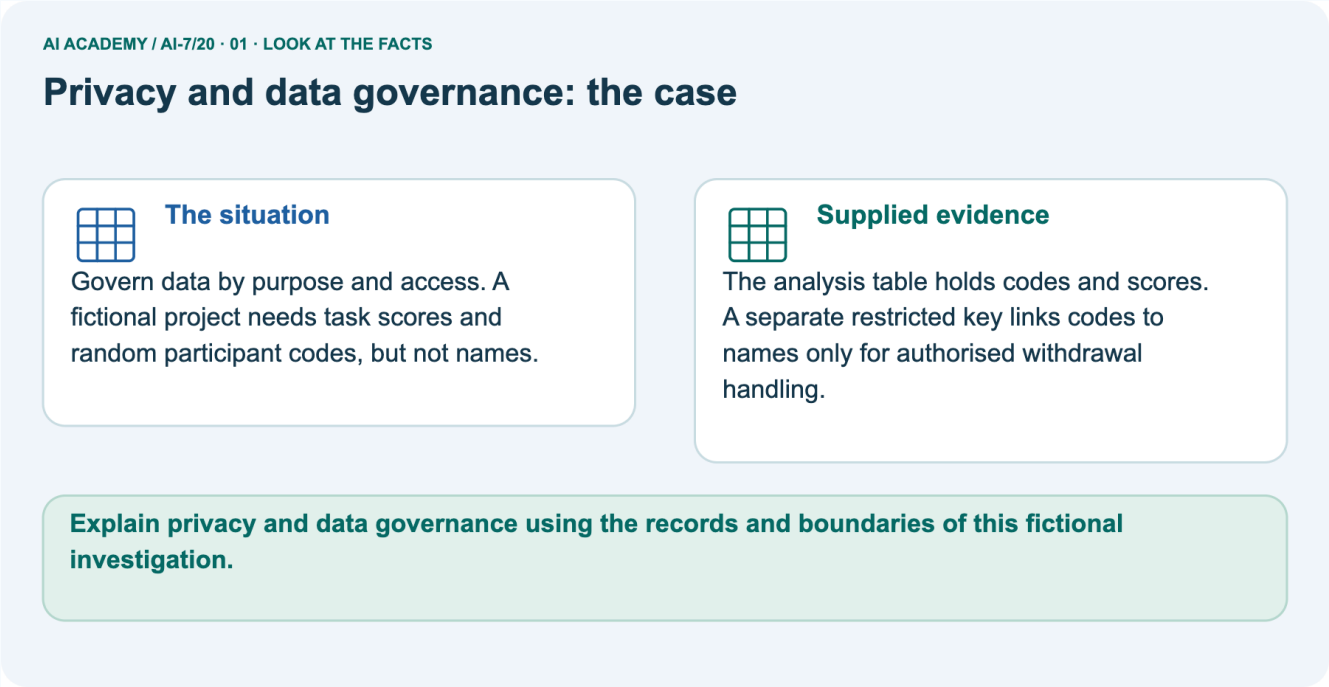
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Privacy and data governance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

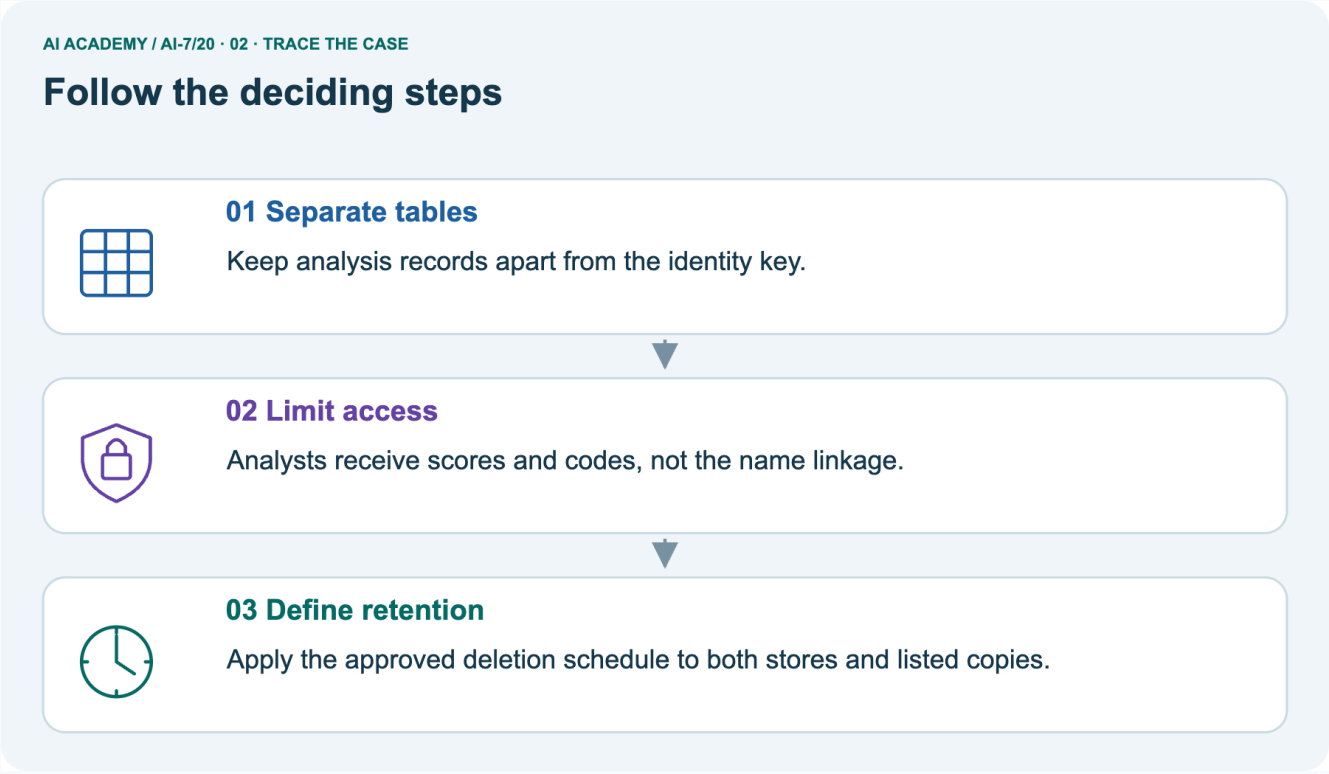
Govern data by purpose and access. A fictional project needs task scores and random participant codes, but not names. Supplied facts: The analysis table holds codes and scores.

A separate restricted key links codes to names only for authorised withdrawal handling.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



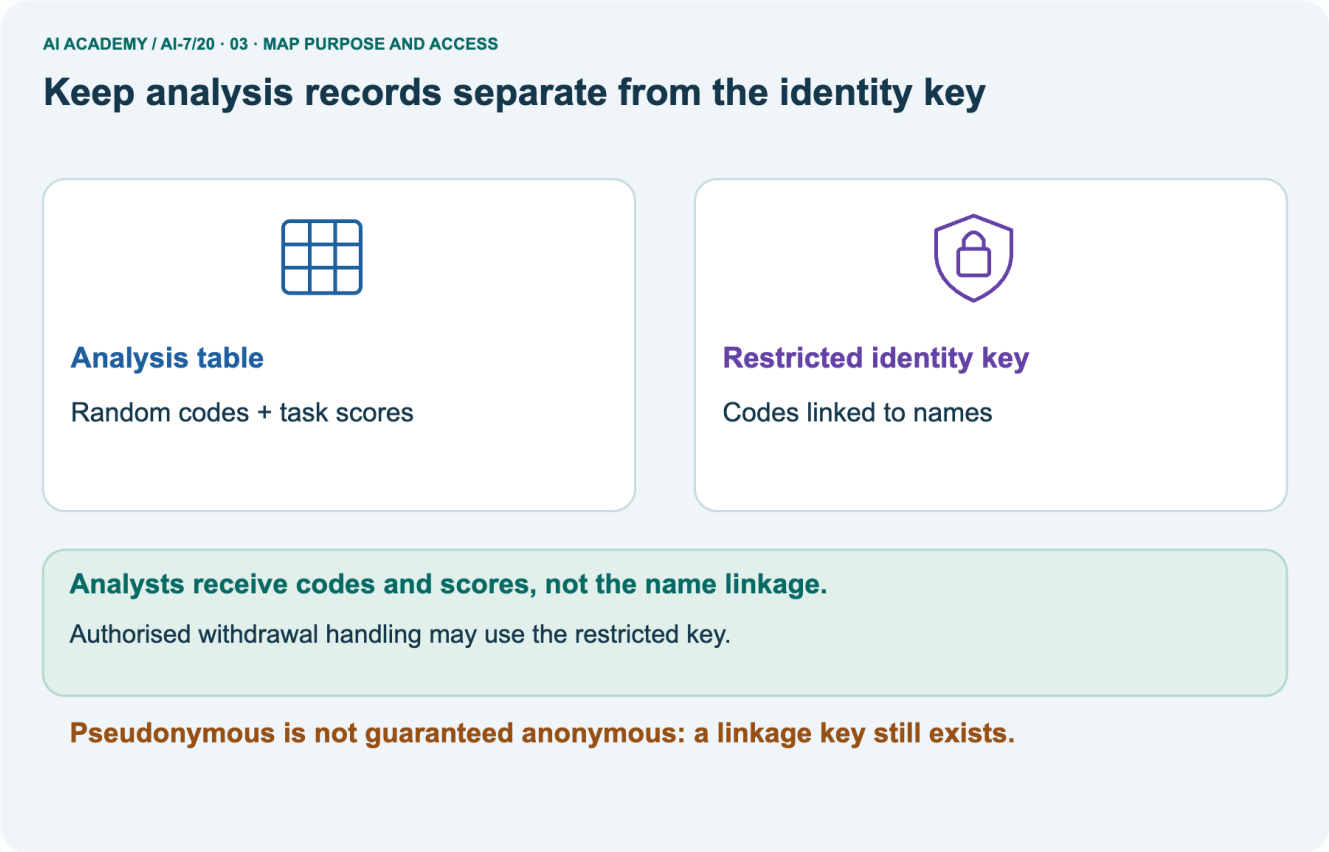
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate tables: Keep analysis records apart from the identity key. → Limit access: Analysts receive scores and codes, not the name linkage. → Define retention: Apply the approved deletion schedule to both stores and listed copies.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep analysis records separate from the identity key



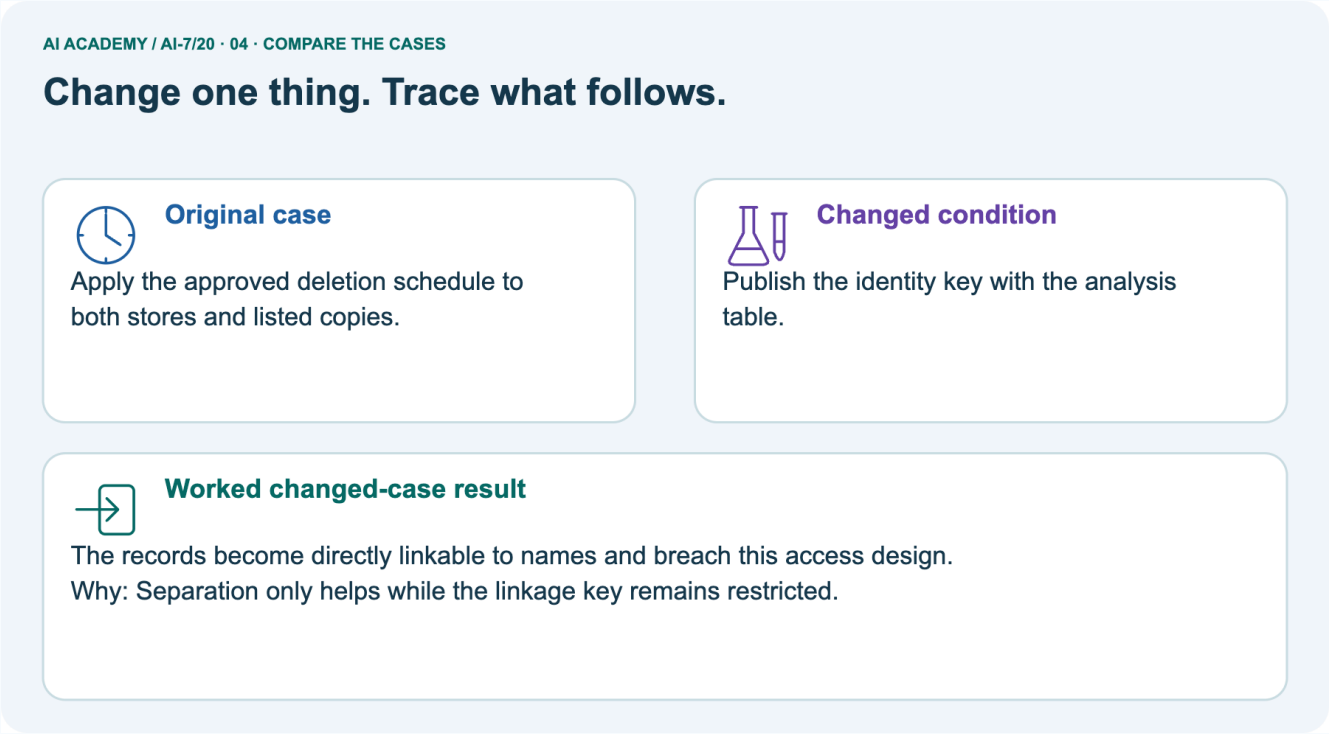
Follow the labelled picture with the matching explanation. Open the vector diagram

Analysts receive random participant codes and scores. A separate restricted key links codes to names only for authorised withdrawal handling. Keeping the key restricted reduces exposure; it does not guarantee anonymity.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

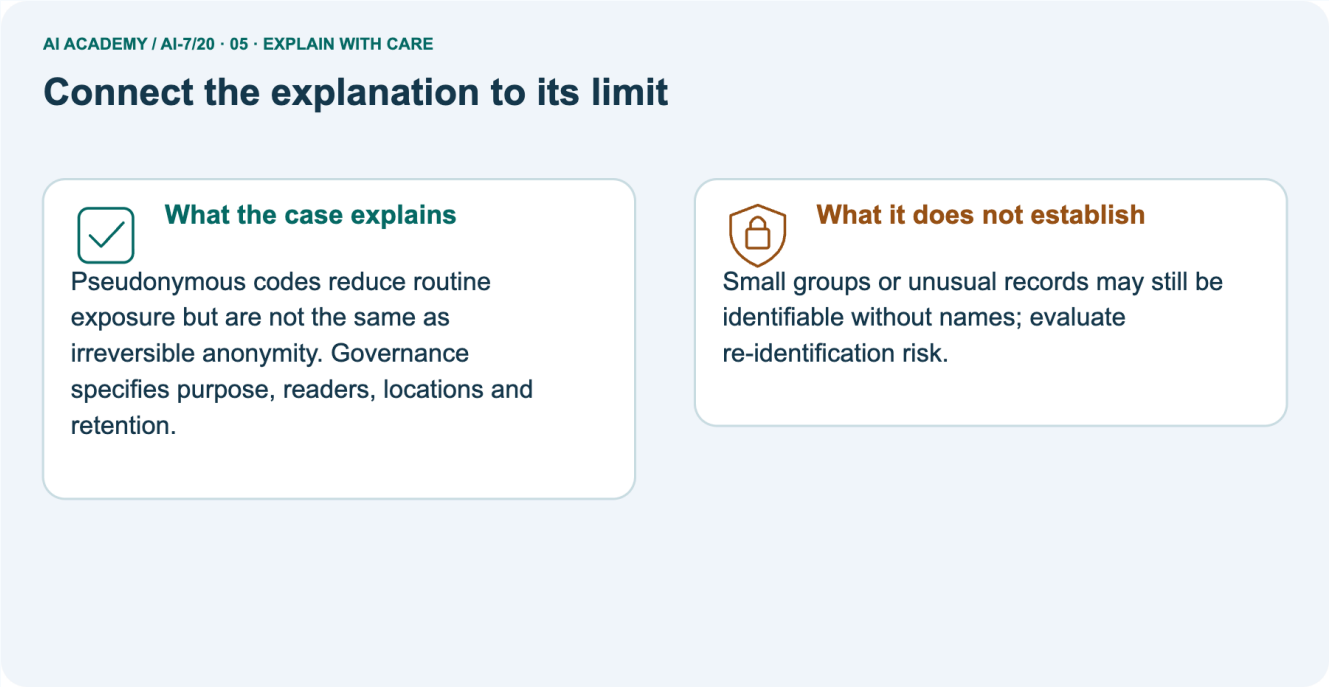
Original case: Apply the approved deletion schedule to both stores and listed copies.
Changed condition: Publish the identity key with the analysis table.

Expected result: The records become directly linkable to names and breach this access design. Reason: Separation only helps while the linkage key remains restricted.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

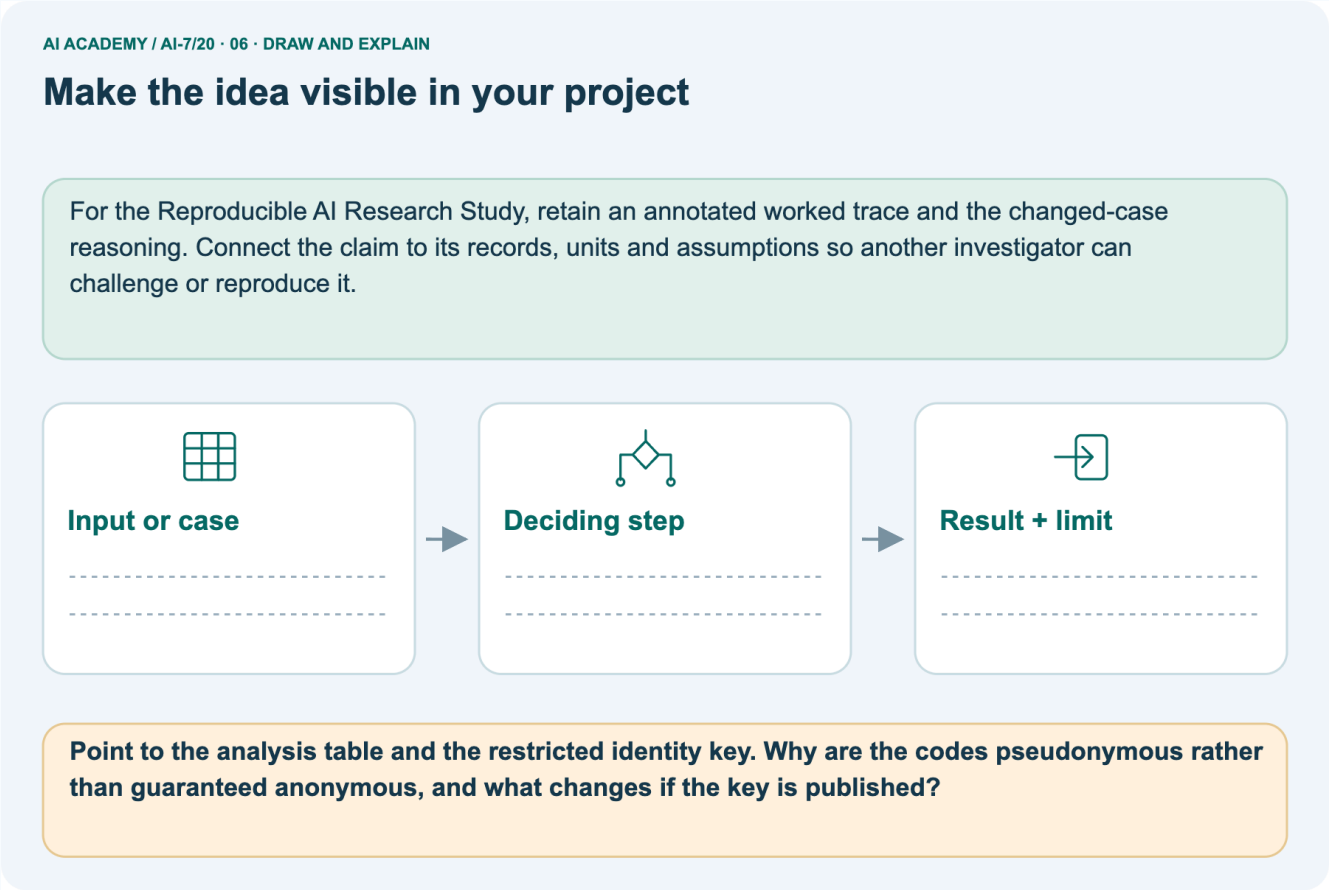
Pseudonymous codes reduce routine exposure but are not the same as irreversible anonymity. Governance specifies purpose, readers, locations and retention.

Limit: Small groups or unusual records may still be identifiable without names; evaluate re-identification risk.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the analysis table and the restricted identity key.

Why are the codes **pseudonym**ous rather than guaranteed anonymous, and what changes if the key is published?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Privacy and data governance: the case

02. Follow the deciding steps

03. Keep analysis records separate from the identity key

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.